Dr. Amit Kumar Bhattacharjee

Contact Information

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Education

- ♣ Ph.D. (Theoretical Physics), **Institute of Mathematical Sciences**, Chennai, India ('04-'10) *[Registered: 01/09/06, Submitted: 28/02/10, Defended: 05/12/11, Awarded: 15/02/13].*
- ♣ M.Sc. (Physics), **Indian Institute of Technology**, Kharagpur, India ('02-'04).
- ♣ B.Sc. (Physics Honours), **B.B. College**, University of Burdwan, India ('99-'02).
- ♣ X^{th} & XIIth, DVC HS School, **W.B.B.S.E.** & **W.B.C.H.S.E.**, India ('97,'99).

Appointment

- Assistant Professor (Stage-II), **Asutosh College**, Kolkata, India ('21 onwards).
- Assistant Professor (Stage-I), **Asutosh College**, Kolkata, India ('17-'21).
- DST-INSPIRE Faculty, **Asutosh College**, Kolkata ('17-'20) & **Indian Institute of Science**, Bangalore, India ('15-'17).
- Visiting Researcher, **Institute of Mathematical Sciences**, Chennai, India ('15, 3 months).
- Assistant Researcher in Applied Mathematics, **Courant Institute**, New York, USA ('13-'15).
- Helmholtz-University Young Investigator, **University of Konstanz**, Germany ('12-'13).
- DLR-DAAD Post Doctoral Fellow, **German Aerospace Center** Köln, Germany ('10-'12).

Professional Courses

- ♮ UGC Sponsored 1-Week Short Term Course on Disaster Management, **UGC-MMTTC, University of Mysore**, Varanasi ($28^{th} October - 03^{rd} November$, 2026).
- ♮ Short Term Program on AI For Teaching & Research (Online), **UGC-MMTTC, Pondicherry University**, Varanasi ($05^{th} - 10^{th} October$, 2026).
- ♮ 13th Refresher Course in Environmental Studies & Sustainable Development (Multidisciplinary)(Online), **UGC-MMTTC, Banaras Hindu University**, Varanasi ($11^{th} - 26^{th} August$, 2025).
- ♮ 1st Online Refresher Course in Physics, **UGC-HRDC, Gujarat University**, Ahmedabad ($7^{th} - 20^{th} September$, 2020).
- ♮ 30th Orientation Course (Online), **UGC-HRDC, Mizoram University**, Aizawl ($28^{th} July - 17^{th} August$, 2020).

Honours & Awards

- § Ranked 6th in ***College Service Examination***, West Bengal State ('17).
- § Work featured in ***NewsRX (Science Letter)*** stating "**Researchers from IISc report findings in Science**" ('17),
- § Awarded ***DST-INSPIRE Faculty*** from *INSA-DST*, Govt. of India ('15-'20).
- § Work selected for **"Francois Naftali Frenkiel Award"** by *Physics of Fluids* ('15) & featured in ***Phys.org*** highlighting "**Mathematicians model fluids at the mesoscale**".
- § ***Research Scientist***, Courant Institute of Math. Sciences, New York University, USA ('13).
- § Work selected for **"Special Topics in Glass Transition"** issue by *J. Chem. Phys.* ('13).
- § ***Helmholtz-University Young Investigator*** at University of Konstanz, Germany ('12).
- § ***DLR-DAAD*** award from German Aerospace Centre Köln, Germany ('10).
- § Ranked 128th in ***Joint Entrance Screening Test*** [**JEST**] ('04).
- § Ranked 117th (95.79%) in ***Graduate Aptitude Test in Engineering***[**GATE**] ('04).
- § Awarded CSIR-JRF & LS in **Joint CSIR-UGC JRF (NET) & LS**, Govt. of India ('04).
- § ***DST-Summer Research*** fellow at SN Bose National Centre for Basic Science, India ('03).
- § ***National Scholarship*** from **Department of Education**, Govt. of India ('03).
- § Ranked 6th in ***Admission Test for M.Sc.***, IIT Kharagpur ('02).
- § ***"University Silver Medal"***, B.Sc. 2nd rank in **University of Burdwan**, India ('02).
- § DVC 1st prize for performance in XIIth Board Examination, **WBCHSE**, India ('99).
- § DVC 2nd prize for **outstanding** performance in X^{th} Board Examination, **WBBSE**, India ('97).

RESEARCH EXPERTISE

Soft Condensed Matter Theory & Computation: ***(a) Field theoretic methods*** (μm-m,μs-hr): (i) Fluctuating hydrodynamics with Projection methods, (ii) hybrid Lattice-Boltzmann method, (iii) Landau-de Gennes energy landscape method, ***(b) Particle based methods*** (pm-nm,ps-ns): (iv) Molecular dynamics simulation, (v) Kinetic monte carlo methods, ***(c) Multiscale methods***: (vi) Dissipative particle dynamics simulations, (vii) High performance computation (HPC).

PEER REVIEWED PUBLICATIONS [*Citations:* **253**, *h-index:* **7**, *i10-index:* **7**, *Scopus ID:* **56556042400**, *ORCID ID:* **0000-0002-1475-743X**, *Web of Science :* **AAB-1030-2020**]

LIQUID CRYSTALS :

√ **A.K. Bhattacharjee**. Controlling motile disclinations in a thick nematogenic material with an electric field, ***Nature (Scientific Reports)***, **8**, 2517 (2018), [citation:**1**, pages:18, ISSN:2045-2322, IF:4.379].

√ **A.K. Bhattacharjee**. Stochastic kinetics reveal imperative role of anisotropic interfacial tension to determine morphology and evolution of nucleated droplets in nematogenic films, ***Nature (Scientific Reports)***, **7**, 40059 (2017), *(Highlighted in Review Articles in "* **2019, 2025** *" & featured in NewsRX "* **(Science Letter)** *")*, [citation:**7**, pages:17, ISSN:2045-2322, IF:4.379].

√ **A.K. Bhattacharjee**. Inhmogeneous Phenomena in Nematic Liquid Crystals, ***Homi Bhabha National Institute*** (2013), [citation:**7**, pages:124, PhD Thesis].

√ **A.K. Bhattacharjee**, Gautam I. Menon and R. Adhikari. Fluctuating dynamics of nematic liquid crystals using the stochastic method of lines, ***J. Chem. Phys.*** **133**, 044112 (2010), [citation:**27**, pages:7, ISSN:1089-7690, IF:3.48].

√ S.M. Kamil, **A.K. Bhattacharjee**, R. Adhikari and Gautam I. Menon. The isotropic-nematic interface with an oblique anchoring condition, ***J. Chem. Phys.*** **131**, 174701 (2009), [citation:**14**, pages:10, ISSN:1089-7690, IF:3.48].

√ S.M. Kamil, **A.K. Bhattacharjee**, R. Adhikari and Gautam I. Menon. Biaxiality at the isotropic - nematic interface with planar anchoring, ***Phys. Rev. E*** **80**, 041705 (2009), [citation:**19**, pages:5, ISSN:2470-0053, IF:2.529].

√ **A.K. Bhattacharjee**, Gautam I. Menon and R. Adhikari. Numerical method of lines for the relaxational dynamics of nematic liquid crystals, ***Phys. Rev. E*** **78**, 026707 (2008), [citation:**35**, pages:10, ISSN:2470-0053, IF:2.529].

COLLOIDAL GLASS :

√ **A.K. Bhattacharjee**. Stress-structure relation in dense colloidal melt under forward and instantaneous reversal of shear. ***Soft Matter (Royal Society of Chemistry)***, **11**, 5697 (2015), [citation:**5**, pages:8, ISSN:1744-6848, IF:3.399].

√ F. Frahsa, **A.K. Bhattacharjee**, J. Horbach, M. Fuchs and Th. Voigtmann. On the Bauschinger effect in supercooled melts under shear: results from MCT and molecular dynamics simulation, ***J. Chem. Phys.*** **138**, 12A513 (2013), *(Appeared in "* **Special Topics in Glass Transition** *")*, [citation:**31**, pages:14, ISSN:1089-7690, IF:3.48].

LIQUIDS & GASES :

√ **A.K. Bhattacharjee**, K. Balakrishnan, A. L. Garcia, J.B. Bell and A. Donev. Fluctuating hydrodynamics of multispecies reactive mixtures. ***J. Chem. Phys.***, **142**, 224107 (2015), [citation:**57**, pages:22, ISSN:1089-7690, IF:3.48].

√ A. Donev, A.J. Nonaka, **A. K. Bhattacharjee**, A. L. Garcia and J. B. Bell. Low Mach Number Fluctuating Hydrodynamics of Multispecies Liquid Mixtures. ***Physics of Fluids*** **27**, 037103 (2015), *(Selected for*

" **Francois Naftali Frenkiel Award** " *& featured in* " **Phys.org** "),
[citation:**49**, pages:35, ISSN:1089-7666, IF:3.521].

COLLEGE PUBLICATIONS :

√ **A.K. Bhattacharjee**. Complexity in the fluctuating hydrodynamical theory of simple fluids. ***The Centurion*** **10**, 105-113 (2025), [citation:**x**, ISSN:2456-4206].

√ **A.K. Bhattacharjee**. Career Progression for a Physics undergraduate. ***Career Guidance: Choices Before You*** **9**, 123-128 (2023), [citation:**x**, ISBN:978-91-966693-8-6].

√ **A.K. Bhattacharjee**. A computational study of the oscillator problems. ***Critical Studies in Science*** **1**, 371-378 (2023), [citation:**x**, ISBN:978-81-966693-7-9].

√ **A.K. Bhattacharjee**. Thrills of Hiking in the Montafon Valley. ***Green Thrills of Travels, Tours and Adventures*** **1**, 46-54 (2023), [citation:**x**, ISBN:978-81-966693-6-2].

√ R. Ray, P. Rudra, **A.K. Bhattacharjee**, S. Chaudhuri, N. Mukherjee and S. Sen. Teaching-Learning during the Pandemic and After: A Multi-Disciplinary Approach. ***The Covid*** **19**, 15-26 (2022), [citation:**1**, ISBN:978-81-956797-1-3].

√ **A.K. Bhattacharjee**. Topological defects in liquid crystal films: formation, annihilation and manipulation. ***The Centurion*** **4**, 105-113 (2019),[citation:**x**, ISSN:2456-4206].

Invited Reviewer

¶ Journal reviewer: **Soft Matter (RSC)**, **Physical Review**, **Reviews of Modern Physics**.
¶ Proposal reviewer of **Netherlands Organisation for Scientific Research (NWO)**.
¶ Editor of **Centurion Teachers Council Journal**, Asutosh College, Kolkata, India.
¶ Biographical interview by **Deutsche Welle** at DLR, Germany.

Teaching / Mentoring Experience

(a) Asutosh College, Kolkata:

2023-25

- Certificate Course on "LaTeX" renamed to **"Scientific Paper Writing Tools"**.
- *PHSA (PHSG)* Courses in CBCS 2019 Curriculum changed to *PHSM (MPHS)* in CCF 2022 Framework.

2022-23

- In Addition to 2021-22: Certificate Course on **"LaTeX"**.
- Year-*III* (PHSA) **Computer Laboratory** (in C) course is abolished.

2021-22

- Sem-*VI* (PHSA) : ***(i)*** **Advanced Classical Dynamics**, ***(ii)*** **Advanced Statistical Mechanics**, & ***(iii)*** **Statistical Mechanics** (Python).
- Sem-*V* (PHSA) : ***(i)*** **Laser & Fiber Optics** (Holography & Nonlinear Optics), & ***(ii)*** **Statistical Mechanics** (Python).
- Sem-*IV* (PHSA) : Mathematical Physics III: ***(i)*** **Special Relativity** & ***(ii)*** **Python**.
- Sem-*III* (PHSA) : ***(i)*** **Thermal Physics**, ***(ii)*** **Mathematical Physics II** (Python) & ***(iii)*** **Scientific Writing** (LaTeX).
- Sem-*II* (PHSA) : **Physical Optics** (Diffraction & Waves).
- Sem-*I* (PHSA) : **Mechanics** (Nonintertial Frame & Fluid Mechanics).
- ♮ Sem-*V* (PHSG) : **Modern Physics** (Special Relativity).
- ♮ Sem-*IV* (PHSG) : **Physical Optics** (Polarization & Diffraction).
- ♮ Sem-*III* (PHSG) : **Scientific Writing** (LaTeX).
- Year-*III* (PHSA) : **Computer Laboratory** (in C).

2020-21

- Sem-*V* (PHSA) : ***(i)*** **Solid State Physics**, ***(ii)*** **Laser & Fiber Optics** (Holography & Nonlinear

Optics), & *(iii)* **Quantum Mechanics** (Python).
- Sem-*IV* (PHSA) : Mathematical Physics III: *(i)* **Special Relativity** & *(ii)* **Python**.
- Sem-*III* (PHSA) : *(i)* **Thermal Physics** & *(ii)* **Mathematical Physics II** (Python).
- Sem-*II* (PHSA) : **Physical Optics** (Diffraction).
- Sem-*I* (PHSA) : **Mechanics** (Nonintertial Frame & Fluid Mechanics).
- ↳ Sem-*V* (PHSG) : **Modern Physics** (Special Relativity).
- ↳ Sem-*I* (PHSG) : **Mechanics** (Oscillations).
- Mentored U.G. student S.Paul (Asutosh College) in a project "Computational study of Conserved & Non-conserved Systems".

2019-20
- Year-*III* (PHSA) : *(i)* **Solid State Physics**, & *(ii)* **Computer Laboratory** (in C).
- Sem-*III* (PHSA) : *(i)* **Thermal Physics** & *(ii)* **Mathematical Physics II** (Python).
- Sem-*II* (PHSA) : **Physical Optics** (Diffraction & Holography).
- Sem-*I* (PHSA) : **Mechanics** (Nonintertial Frame & Fluid Mechanics).
- ↳ Sem-*II* (PHSG) : **Electrodynamics** (Induction & Maxwell's Equation).
- ↳ Sem-*I* (PHSG) : **Mechanics** (Oscillations & Elasticity).

2018-19
- Sem-*II* (PG) (ENVS) : **Environment & Energy**.
- Year-*III* (PHSA) : *(i)* **Solid State Physics**, & *(ii)* **Computer Laboratory** (in C).
- Year-*II* (PHSA) : **Thermal Physics II**.
- Sem-*I* (PHSA) : *(i)* **Mechanics** (Nonintertial Frame & Fluid Mechanics), & *(ii)* **Mathematical Physics I** (Python).
- ↳ Year-*III* (PHSG) : *(i)* **Computer Laboratory** (in C) & *(ii)* **Communication Theory**.
- ↳ Sem-*I* (PHSG) : **Mechanics** (Oscillations & Elasticity).
- Mentored Ph.D. student S. Anand (IISER Bhopal) in a project "Electrically Driven Droplets".

2017-18
- Year-*III* (PHSA) : *(i)* **Solid State Physics**, & *(ii)* **Computer Laboratory** (in C).
- Year-*II* (PHSA) : **Thermal Physics II**.
- Year-*I* (PHSA) : **Thermal Physics I**.
- ↳ Year-*III* (PHSG) : *(i)* **Computer Laboratory** (in C) & *(ii)* **Communication Theory**.
- ↳ Year-*I* (PHSG) : **Mechanics** (Waves & Oscillations).

(b) Indian Institute of Science, Bangalore:

2015-16
- Mentored a Ph.D. student (P.J.Bhuiyan) in a project "Emergent Structures in Colloidal Membranes" (Fall).
- Mentored a U.G. student (A.Shetty, IIT Bombay) in a project "Nematic Rheochaos in 2D" (Summer).

(c) Universität Konstanz, Konstanz:

2012-13
- Mentored M.Sc. student (M.Everts) towards "**Ausarbeitung**" in the course **Materie und Ordnung** (Spring).
- Bilingual Teaching Assistant/Grader (German & English) in the course **Classical Field Theory** (Fall).

EXAMINER/PAPER-SETTER/SCRUTINIER/REVIEWER (UNIVERSITY OF CALCUTTA)
*[**E**: Examiner, **S**: Scrutinier, **R**: Reviewer, **P**: Paper-Setter, **M**: Moderator, **HE**: Head-Examiner, **In**: Internal, **Tu**: Tutorial, **Th**: Theory]*

Examiner,Scrutinier,Head-Examiner, Paper-Setter (Theory)

2025
- Sem-*VI* (PHSA) : Paper-DSE *A2*(*B*) (Advanced Dynamics){In,Tu,Th}[**E**=15,**S**=15].
- Sem-*IV* (PHSM) : Paper-DSCC8 (Classical Mechanics & Special Theory of Relativity){In,Th}[**E**=28, **S**=15].
- ¶ Undisclosed Competitive Examination [**P**,**E**=100,**S**=200].

2024
- Sem-*VI* (PHSA) : Paper-DSE *A2*(*B*) (Advanced Dynamics){In,Tu,Th}[**E**=35,**S**=35].

- Sem-III (PHSA) : Paper-SEC A-1 (Scientific Writing (LaTeX)){In,Tu,Th}[**E**=32].
- Sem-III (MPHS) : Paper-CC-3 (Waves & Optics){Th}[**HE**=300].

2023

- Sem-VI (PHSA) : Paper-DSE $A2(B)$ (Advanced Dynamics){In,Tu,Th}[**E**=16,**S**=3].
- Sem-V (PHSA) : Paper-DSE $A1(B)$ (Laser & Fiber Optics){In,Tu,Th}[**E**=33, **S**=31].
- Sem-III (PHSA) : Paper-CC6 (Thermal Physics){In,Th}[**E**=22, **S**=23].
- Sem-III (PHSA) : Paper-SEC A-1 (Scientific Writing (LaTeX)){In,Tu,Th}[**E**=22].
- Sem-II (PHSA) : Paper-CC4 (Waves & Optics){In,Th}[**E**=25, **S**=xx].

[⋈] Sem-III (PHSG) : Paper-SEC A-1 (Scientific Writing (LaTeX)){In,Tu,Th}[**HE**, **E**=14].

¶ Undisclosed State University [**P**].

2022

- Sem-VI (PHSA) : Paper-DSE $A2(B)$ (Advanced Dynamics){In,Tu,Th}[**E**=18, **S**=10].
- Sem-V (PHSA) : Paper-CC12 (Solid State Physics){In,Th}[**E**=34, **S**=34].
- Sem-V (PHSA) : Paper-DSE $A1(B)$ (Laser & Fiber Optics){In,Tu,Th}[**E**=48, **S**=48].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III){In,Th}[**E**=42, **S**=42].
- Sem-III (PHSA) : Paper-CC6 (Thermal Physics){In,Th}[**E**=46, **S**=46].
- Sem-III (PHSA) : Paper-SEC A-1 (Scientific Writing (LaTeX)) [**E**=43].
- Sem-I (PHSA) : Paper-CC2 (Mechanics){In,Th}[**E**=60, **S**=60].
- Year-III (PHSA) : Paper-6 (Solid State Physics) [**E**=9, **S**=9].
- Year-I (PHSA) (1+1+1 System Syllabus 2010-11) : Paper-I (Mathematical Physics).

♮ Sem-V (PHSG) : Paper-DSE $A(2)$ (Modern Physics){In,Th}[**E**=38, **S**=38].

♮ Sem-IV (PHSG) : Paper-$CC4/GE4$ (Optics){In}[**E**=142].

♮ Sem-I (PHSG) : Paper-$GE/CC1$ (Mechanics){In,Th}[**E**=101, **S**=101].

¶ Undisclosed State University [**P**].

2021

- Sem-VI (PHSA) : Paper-DSE $A2(B)$ (Advanced Dynamics) {In,Tu,Th}[**E**=34, **S**=34].
- Sem-VI (PHSA) : Paper-DSE $B2(B)$ (Non-eq. Statistical Mechanics){In,Tu,Th}[**E**=34, **S**=34].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III){In,Th}[**E**=47, **S**=47].
- Sem-II (PHSA) : Paper-CC4 (Waves & Optics){In,Th}[**E**=42, **S**=42].

♮ Sem-IV (PHSG) : Paper-$CC4/GE4$ (Waves & Optics) [**E**=154, **S**=154].

- Year-III (PHSA) : Paper-6 (Nuclear & Solid State Physics) [**E**=9, **S**=9].
- Sem-V (PHSA) : Paper-DSE $A1(B)$ (Laser & Fiber Optics){In,Tu,Th}[**E**=34, **S**=34].
- Sem-V (PHSA) : Paper-CC12 (Solid State Physics){In,Th}[**E**=34, **S**=34].
- Sem-III (PHSA) : Paper-CC6 (Thermal Physics){In,Th}[**E**=43, **S**=43].
- Sem-I (PHSA) : Paper-CC2 (Mechanics){In,Th}[**E**=xx, **S**=xx].

♮ Sem-V (PHSG) : Paper-DSE $A(2)$ (Modern Physics){In,Th}[**E**=46, **S**=46].

¶ Undisclosed State University [**P**].

2020

- Year-III (PHSA) : Paper-6 (Nuclear & Solid State Physics) [**E**=80, **S**=80].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III){In,Th}[**E**=34, **S**=34].
- Sem-III (PHSA) : Paper-CC6 (Thermal Physics){In,Th}[**E**=44, **S**=47].
- Sem-II (PHSA) : Paper-CC4 (Waves & Optics){In,Th}[**E**=48, **S**=48].

♮ Year-II (PHSG) : Paper-$IIIA$ (Optics Electronics, Modern Physics) [**E**=71, **S**=71].

2019

♮ Year-I (PHSG) : Paper-I (Math.Methods, Geom.Optics & Electronics) [**E**=20, **S**=20].

♮ Sem-II (PHSG) : Paper-$GE/CC2$ (EM Theory) [**E**=11, **S**=16].

♮ Sem-I (PHSG) : Paper-$GE/CC1$ (Mechanics) [**E**=37, **S**=49].

2018

¶ Sem-II (**PG**) (ENVS) : Paper-$ENVC-24$ (Energy & Environment), University of Calcutta[**E**=**P**=25].

- Year-III (PHSA) : Paper-VI (Nuclear & Solid State Physics) [**E**=33, **S**=37].

♮ Year-III (PHSG) : Paper-IVA (Thermodynamics, Electronics) [**E**=55, **S**=53].

♮ Year-II (PHSG) : Paper-$IIIA$ (Optics, Electronics, Modern Physics) [**E**=44, **S**=49].

2017

- Year-III (PHSG) : Paper-IVA (Thermodynamics, Electronics) [**E**=31, **S**=50].
- Year-II (PHSG) : Paper-$IIIA$ (Optics, Electronics, Modern Physics) [**E**=50, **S**=71].
- Year-I (PHSG) : Paper-I (Mechanics, GPM, Oscillations, Optics) [**E**=67, **S**=76].

Internal,External Examiner & Scrutinier (Computer)

2025

- Sem-V (PHSM) : DSC-7 (Asutosh College) (Mathematical Physics II)[**E**=24].
- Sem-III (PHSM) : DSC-4 (Asutosh College) (Mathematical Physics I)[**E**=26].

2024

- Sem-V (PHSM) : DSCC-4 (Maulana Azad College) (Mathematical Physics)[**E**=34].
- Sem-V (PHSA) : Paper-CC12 (Jogamaya Devi College) (Statistical Mechanics)[**E**=22].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III)[**E**=27].

2023

- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III)[**E**=45].
- Sem-V (PHSA) : Paper-CC12 (Jogamaya Devi College) (Statistical Mechanics)[**E**=47].
- Sem-III (PHSA) : Paper-CC5 (Mathematical Physics-II)[**E**=27].

2022

- Sem-IV (PHSA) : Paper-CC8 (Jogamaya Devi College) (Mathematical Physics-III)[**E**=12].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III)[**E**=42].
- Sem-VI (PHSA) : Paper-CC12 (Statistical Mechanics)[**E**=48].
- Sem-III (PHSA) : Paper-CC5 (Mathematical Physics-II)[**E**=43].

2021

- Sem-VI (PHSA) : Paper-CC14 (Statistical Mechanics)[**E**=34].
- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III)[**E**=47].
- Sem-III (PHSA) : Paper-CC5 (Mathematical Physics-II)[**E**=46].
- Year-III (PHSA) : Paper-$VIIIB$ (Computer Laboratory)(in C)[**E**=9].

2020

- Sem-IV (PHSA) : Paper-CC8 (Mathematical Physics-III)[**E**=34].
- Year-III (PHSA) : Paper-$VIIIB$ (Computer Laboratory)(in C)[**E**=80, **S**=80].
- Sem-I (PHSA) : Paper-$CC1$ (Heritage College)[**E**=19].

2018

- Sem-I (PHSA) : Paper-$CC1$ (Heritage College)[**E**=19, **S**=120].

Internal,External Examiner & Scrutinier (Experiment)

- Sem-I (PHSA) : Paper-DSCC1 (Mechanics) (2024) [**E**=18].
- Sem-II (PHSA) : Paper-CC3 (Jogamaya Devi College) (Electricity Magnetism) (2022) [**E**=12].
- Sem-I (PHSG) : Paper-GE/CC1 (2020) [**E**=96].
- Year-II (PHSG) : Paper-$IIIB$ (Optics Electronics, Modern Physics) (2020) [**E**=71].
- Year-II (PHSG) : Paper-IIA (2019) [**E**=92].
- Year-II (PHSG) : Paper-IIA (2018) [**E**=105].
- Year-III (PHSG) : Paper-IVA (2017) [**E**=100].

Invited Speaker

- **Student's Week**, Asutosh College, India (January 2022).
- **Centre for Computational & Data Sciences**, IIT Kharagpur, India (December 2018).
- **Complex Fluids - CompFlu-2017**, IIT Madras, India (December 2017).
- **Complex Fluids - CompFlu-2016**, IIIT Hyedarabad, India (December 2016).
- Institute Seminar, RRI Bangalore, India (September 2016).
- **Theoretical Science Unit, JNCASR Bangalore, India (March 2016)**.
- Thematic Unit of Excellence in Computational Material Science, SNBNCBS Kolkata, India (January

2016).

- ♠ Department of Physics, IIT Delhi, India (January 2016).
- ♠ 3rd **Soft Matter Young Investigator Meet, Pondicherry, India (December 2015)**.
- ♠ Journal Club, The Institute of Mathematical Sciences, Chennai, India (July 2015).
- ♠ Department of Physics, IISER Mohali, India (April 2015).
- ♠ **School of Physical Sciences, JNU New Delhi, India (April 2015)**.
- ♠ Department of Physics, IISER Bhopal, India (April 2015).
- ♠ Workshop Bartholomäberg, Vorarlberg, Austria (August 2012).
- ♠ **Konstanzer Kolloidal Klub, Universität Konstanz, Konstanz, Germany (June 2012)**.
- ♠ Fachbereich Physik, Universität Konstanz, Konstanz, Germany (February 2012).
- ♠ Institut für Theoretische Physik, Heinrich-Heine-Universität Düsseldorf, Germany (October 2011).
- ♠ Institut für Materialphysik im Weltraum, DLR Köln, Germany (April 2011).
- ♠ **Journal Club, The Institute of Mathematical Sciences, Chennai, India (April 2009)**.
- ♠ Mahabaleswar Seminar on Modern Biology, TIFR, Mumbai, India (January 2008).

Attended Conferences/ Workshops / Webinars

- One Day Workshop on Physics Syllabi under CCF 2022, **Scottish Church College** Kolkata (16th*October*, 2023).
- © One Day Seminar to celebrate "**165th Birthday of Sir J.C. Bose**" at IIEST, Shibpur (Nov 2023).
- Two Day National Webinar on Fundamental Physics, **Asutosh College** Kolkata ($4^{th} - 5^{th}$*October*, 2021).
- National Webinar on Emergent Phenomena: When More Is Different, **Raja Peary Mohan College** Hooghly (20th*September*, 2021).
- Six Day Workshop on Nonlinear Dynamics, **Bangabasi College** Kolkata (28th*June* – 3rd*July*, 2021).
- One Day Webinar on Quantum mechanics with Python, **Fakir Chand College** Diamond Harbour (2nd*July*, 2021).
- Faculty Development Program on Quantum Mechanics in Python, **Prabhu Jagatbandhu College** Howrah (19th*February*, 2021).
- National Webinar on Some Selected Topics on DSE Physics Course Under CBCS, CU, **Maulana Azad College** Kolkata ($23^{rd} - 24^{th}$*September*, 2020).
- National Webinar on Fundamental Physics, **Asutosh College** Kolkata ($27^{th} - 28^{th}$*August*, 2020).
- Web Based Workshop on Teaching Physics at the UG & PG Level using Python, **Victoria Institution** & **UGC-DAE CSR** Kolkata ($6^{th} - 10^{th}$*July*, 2020).
- One Day Seminar-cum-Workshop on Python computing: Some Applications in Mathematical Physics, **Basanti Devi College** & **Nabagram Hiralal Paul College** Kolkata (28th*February*, 2020).
- Two Day Faculty Development Program on Python, **Behala College** Kolkata ($3^{rd} - 4^{th}$*May*, 2019).
- One Day Seminar on Basic Sciences, **RSA-IACS** & **Asutosh College** Kolkata (4th*February*, 2019).
- One Day Workshop on CBCS-Physics Syllabus, **Bangabasi College** Kolkata (7th*May*, 2018).
- One Day Seminar on Sister Nivedita, **Asutosh College** Kolkata (31th*October*, 2017).
- Indian Statistical Physics Community Meeting, ICTS Bangalore, India (2016).
- Growing Length Scale Phenomena, JNCASR Bangalore, India (2015).
- **Kurt Binder honorary workshop, Johannes Gutenberg-Universität Mainz, Germany (2012)**.
- SimBioMa2011, Universität Konstanz, Konstanz, Germany (2011).
- School on Nonlinear Response to Vitrification, Universität Konstanz, Konstanz, Germany (2011).
- Glastag, Universität Marburg, Marburg, Germany (2011).
- 8th **Liquid Matter Conference, Universität Wien, Vienna, Austria (2011)**.
- SERC School cum Symposium on Rheology of Complex Fluids, IIT Madras, India (2010).
- **Disorder, Complexity & Biology II, BHU Varanasi, India (2009)**.
- The Interface of Life, IIT Madras, India (2008).
- School on Understanding Molecular Simulation, JNCASR Bangalore, India (2007).
- Assembly Organization & Propulsion in Complex Systems, IIT Madras, India (2007).
- SERC School on Nonlinear Dynamics & Pattern Formation, IACS Kolkata, India (2006).
- Common Trends in Traffic: Physical & Computational Models in Transportation Engineering & Biological Sciences, IIT Kanpur, India (2006).
- Discussion Meeting on Statistical Physics, Vardanahalli, India (2005).

Experiencial Learning / Outreach Activity

- © Invited Judge of "**Chhatra Yuba Bigyan Mela**" at Beltala Girls High School, Kolkata (Sept 2019).

- © Conducted Heat & Annual Sports "**Krira**", Asutosh College (Dec 2019, Jan 2019 & Feb 2018).
- © **Seminar on "***Computational Science***"** at PG Department of Physics, B.B.College, Asansol (Dec 2009).
- © Question-Answer session with students of Xth std. at DVC High School, Maithan (April 2015).

EXTRACURRICULAR ACTIVITY

- (⋈) **Visharad (5th year)** on Hawaiian Guitar, Nikhil Bharat Sangeet Samiti, Kolkata, 1999.
- (⋈) **Visharad (5th year)** in Art, Pracheen Kala Kendra, Chandigarh, 1999.
- (⋈) **Nature Photography & Birding**.
- (⋈) Globetrotter's Travelogue.